

Cellular environmental issues

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There has been rapid growth of cellular telephone systems throughout the world over the past few years. This has led to increased public awareness of possible electromagnetic compatibility (EMC) problems, fears about the effects on health and the visual impact of base station radio antennas on the surrounding environment. These three aspects are felt to be the main environmental issues and potential barriers to future exploitation of cellular radio.

The EU EMC Directive (mandatory from 1.1.96) stipulates equipment immunity standards which will go some way to alleviating interference problems, but cellular growth itself is likely to have the most marked effect on all cellular environmental issues, because radio transmission power levels will be reduced as traffic capacity is increased through the use of smaller cells and less obvious antennas. This paper presents work performed at BT Laboratories on cellular radio EMC modelling of the GSM system, current guidelines on exposure to radio frequency transmissions, and the likely impact of evolving from large to small cells.

1. Background

Radio has been used for communications from the beginning of this century. It has always had the potential for causing interference to other electrical equipment, has raised concerns about health and its antennas have had a visual impact on the landscape. The past few years have seen rapid growth in the use of cellular telephone systems world-wide and this is the first time that the mass of people have been able to use radio transmitters first hand.

- Licensing of radio services — who has rights?

The issue of a licence and right to operate radio transmitters is sometimes at odds with the individual who has bought or uses equipment and feels that he has the right to expect that its operation is not interfered with by others. The lack of statutory electrical equipment immunity limits has made this situation worse in the past, but the mandatory introduction of the Electromagnetic Compatibility (EMC) Directive 89/336/EEC has improved matters. This document sets out the requirements for the control of radiated and conducted emissions and immunity, including ESD (electrostatic discharge), mains disturbances, surges and lightning induced effects.

The EMC directive came into force on 1 January 1992 but an amended directive created a 4-year transition period ending on 31 December 1995. Until 1996, all products should have met national standards; however, some of these were lacking in respect of sufficient interference immunity standards. This situation

coincided in time with extremely rapid growth of the GSM system, exacerbating the issue of GSM interference to other electrical and electronic equipment.

Most types of electrical and electronic equipment are within the scope of the directive, but there are exceptions such as military equipment, large fixed installations like power stations and oil rigs, and simple electrical components. Some products are required to meet the requirements of specific directives, e.g. toys, medical equipment, telecommunications terminal equipment and motor vehicles. In these cases the EMC requirements specified in each directive take precedence but if these are not specified, the EMC directive applies. This makes it difficult to generalise about the possibility of a particular radio signal field strength interfering with any type of equipment and means that the different immunity standards associated with equipment have to be considered.

- Cellular interference can be caused anywhere

Cellular by design, is a system for mobile users, and this has meant that it has the potential for causing interference to a wide variety of electrical equipment in many locations. This potential for causing interference has been increased by the widespread use of hand-held transceivers which can be used almost anywhere. The use of 'pulsed' digital radio transmissions, such as used

by the GSM system have also increased this potential for interference, although all types of radio systems have the capability of causing interference.

- Health has been considered

Guidelines, setting exposure limits for people to radio wave energy, have been in existence for many years and are not exceeded by using current cellular hand-held transceivers. These guidelines are based upon the heating of body tissue, but do not cover athermal effects. As a result, some fears about athermal effects have been publicised, despite there being no conclusive evidence to back this up.

Health fears have sometimes been mixed with fears about interference to medical electronics, such as hearing aids and heart 'pacemakers' which is an electrical interference issue, although the fears and experience of those concerned are just as important.

- Cellular antennas have become a feature of all urban and rural environments

The visual impact of cellular base station antennas has perhaps received the widest adverse publicity of all environmental issues. The issue has been very marked in the UK because of the rapid network 'roll-out' of four competing cellular operators all trying to retain and attract new customers to their networks. All have their own base stations and do not usually share antennas or base site locations, thereby increasing the effect up to fourfold. Licensing separate operators and encouraging competitiveness is good for pricing, but is an inefficient way to cover the same geographical areas.

Base stations have to be placed where they can provide network access to customers, which means that most are placed in urban and suburban areas, although there are others in rural areas providing continuity of coverage along linking roads and railways. There is an intrinsic issue of attempting to provide good service while not affecting the look of buildings or sky line, with antennas and towers.

This paper discusses what are felt to be the 'top three' environmental considerations at this time, i.e. interference to electrical equipment from GSM radio transmissions, health issues and the visual impact of cellular base station towers and antennas.

2. Introduction

All radio services have the potential for causing interference, raising health fears and having a visual impact on the environment. The rapid growth of world-wide use of GSM cellular systems, which uses pulse type TDMA (time division multiple access) transmission, has raised the level of public concern.

Much work has been performed at BT Laboratories (BTL) on domestic electrical equipment and hearing aid immunity from GSM transmissions, while the susceptibility of medical electronic equipment such as heart pacemakers and work on health issues has been carried out elsewhere. Although the visual impact of base station towers and antennas is not within the remit of BT Laboratories, it is interesting to comment on typical installations and to explore how the growth and evolution of cellular networks will actually reduce the visual impact of such installations.

This paper discusses the potential for interference from GSM transmissions and presents a model which has been used to estimate the amount of interference a hearing aid user may receive during the course of a normal day. The same modelling technique could also be applied to other equipment.

Health issues are also briefly discussed in the context of current exposure guidelines, and references to other work which is in the domain of the medical fraternity.

Finally, the growth and evolution of GSM systems to even smaller outdoor and indoor cells with lower transmitted power requirements is discussed. This presents a choice of technologies, with a rather blurred distinction between high-bandwidth office systems such as may be provided with low transmitted power DECT (digital European cordless telephony) and wireless LAN (local area network) technologies and the lower bandwidth, but at present higher power, GSM systems.

3. GSM radio interference

Although cellular base station sites generally transmit much more radio frequency power than mobiles (customer terminals), the antennas are usually set up to radiate away from the building or other structure they are mounted on. This means that the radiated signal will be very much reduced in power when it intercepts an item of electronic equipment susceptible to interference in another building or at street level many yards from the antenna. There may be exceptions to this, such as in a dense urban environment where an antenna may point into a window of a close building, but even so the signal strength will be much lower than would be experienced from a mobile at very short range. Indeed it has been found that interference to most electronic equipment from GSM mobile transmissions is

not perceptible at more than 5 m, therefore this discussion is limited to GSM interference emanating from mobiles.

Interference takes place when the operation of another piece of equipment is impaired. At the lowest level this may be a buzz, say only just perceived on a 'personal stereo', whereas at the highest level, the safety of human life may be risked, such as by operating a mobile within an aircraft.

All radio frequency transmissions may cause interference, but the pulse-like nature of a TDMA GSM transmission, where the radio frequency carrier is switched off and on abruptly, is more readily demodulated by non-linear components within electronic equipment, than older analogue FM transmissions where the carrier is at a relatively constant amplitude.

The normal RF envelope of the GSM mobile radio transmission is shown in Fig 1.

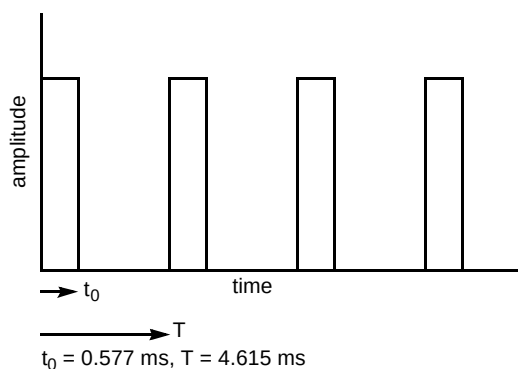


Fig 1 GSM mobile transmitted RF envelope.

There are strong spectral audio components of approximately 200 Hz and 2 KHz caused when the transmitted waveform may be demodulated by a nonlinear component. The pulse rate will be different when the mobile is operating in DTX (discontinuous transmission) mode, which is used to free channel usage when no speech is present, or on other events such as a 'location update' or being 'paged' — however, these are less frequent.

3.1 GSM interference work at BTL

A substantial body of work has been performed at BTL on the GSM interference issue over the past five years. It was realised early on that the nature of GSM transmission could lead to potential interference problems, and thus work was initiated to investigate how it would manifest itself on BT's products and customers. Clearly there could have been a dichotomy of interests, between mobile and fixed line customers; however, a responsible attitude prevailed with both interests being fully involved in the effort to understand the possible interference problems that may be encountered, and also to discuss these findings publicly

within European technical fora attended by representatives from medical and disadvantaged groups such as the hearing impaired. BT also directly set up interference tests with representatives from the hearing impaired community through its community relations department programme, 'Action for Disabled Customers'.

BTL work was taken up by the ETSI group responsible for the GSM radio interface design and incorporated in an 'ETR' [1], with a view to building a body of information on the nature and extent of the GSM EMC problem.

3.2 Interference to domestic appliances

The first work was performed in an anechoic (reflection free) chamber at BTL and investigated the immunity of a range of domestic electronic appliances to GSM-like transmissions. This equipment included a television, personal stereos, a domestic wireless, a video cassette recorder, a hi-fi system and also included hearing aid users [2].

The results showed that significant interference was experienced on some of the items at short range, which were built before there was an immunity standard in place for 900 MHz transmissions. The tests on hearing aids showed that both types tested received interference, but that it was quite directional and depended to some extent on the deafness of the user. It also took place in both amplifying and inductive modes.

The key result from the laboratory hearing aid immunity tests was that perceptible interference was heard when the unit was subjected to a field strength of 4 V/m in some orientations.

3.3 Interference to wired telephones

A more recent study [3] was performed on a selection of wired telephone instruments which were tested for susceptibility to GSM interference from both class 2 (8 W) and class 4 (2 W) mobiles. This was performed in a typical office environment with the interference assessed using the following subjective criteria:

- 0 no interference,
- 1 just perceptible on peaks (of signal strength),
- 2 just perceptible on troughs (some interference at 1 the time),
- 3 intelligible but with loud interference,
- 4 intelligible with difficulty (loud interference),
- 5 no intelligibility (speech distortion and very loud interference).

The results can be presented in two forms which give a simple means of assessing interference:

- a line graph showing the performance rating against distance for the first three samples at both mobile power classes, giving an indication of the level of interference likely to be experienced at a given distance,
- a bar chart showing a comparative performance rating for each instrument when subjected to interference from a GSM mobile — this rating is derived from the product of interference assessment and instrument to mobile separation for comparison purposes.

Examples of these evaluation techniques are shown in Figs 2 and 3.

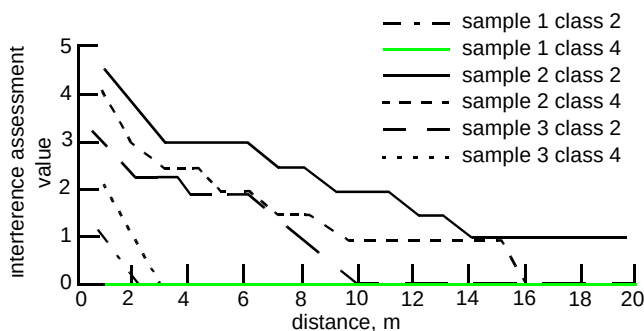


Fig 2 Wired telephone susceptibility to GSM interference.

4. A GSM interference model

The work on domestic appliances and, in particular, the immunity testing of hearing aids showed that the perceived interference depended to a large extent on the attitude of the appliance under test and the GSM source. It has also to be noted that the experiment was carried out under controlled radio conditions within an anechoic chamber and that different conditions would exist within and outside buildings. In addition, a common strategy for maximising spectral efficiency of a cellular system is to minimise the

power transmitted from a mobile. This means that only those mobiles operating at the periphery of cells and possibly some of those within buildings will be operating at full power. Thus the probability of receiving severe GSM interference for a hearing aid user would be considerably reduced. It was decided to model this scenario [4] to estimate the probability of interference which might be encountered by a typical hearing aid user during the course of a day.

4.1 Interference modelling assumptions

The following assumptions were made when modelling this scenario [4]:

- perceptible hearing aid audible interference is heard when subject to a field strength of 4 V/m in some orientations,
- as it has been assumed that most interference would be perceived in an urban rather than a rural environment, where there is a relatively high density of users who are more likely to come in close contact with the hearing aid user, a typical cell size of 2 km radius was chosen, illuminated by a power class 4 transmitter (40 W), with the height of this transmitter chosen to be 62 m, 2 m above the roof tops of typical tall urban buildings,
- all cells would be operating at 50% of their traffic capacity,
- there would be an even mix of class 2 (8 W peak) vehicle-mounted and transportable phones as well as class 4 (2 W peak) hand-held units (this assumption was correct at the time of this work, but the mix has subsequently changed to the major part of the GSM phone population being hand-held (class 4), and this is also the case when used in special vehicle mounts, the model thus overestimating now the level of interference),
- people are evenly distributed within the cell,

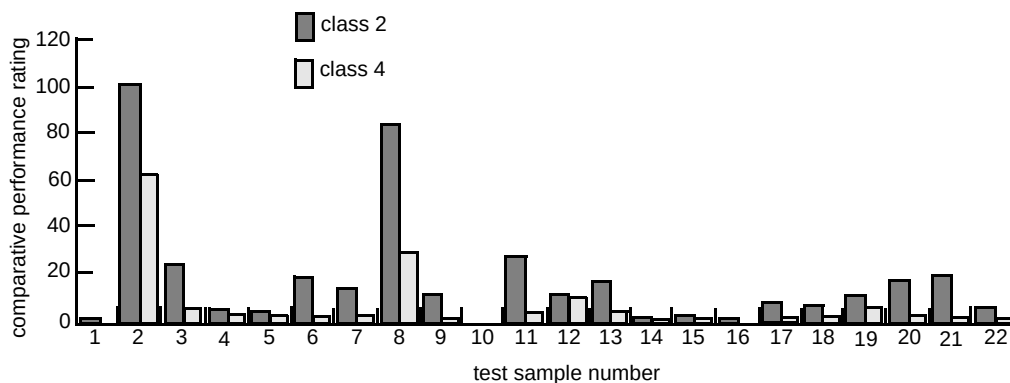


Fig 3 Wired telephone susceptibility to GSM interference from class 2 and class 4 mobiles.

- vehicle-mounted phones are located on three concentric rings within a cell and are distributed in the ratio of their distance from the base station (this was found to be a good approximation to an even distribution of mobiles),
- the number of hearing aids in the UK is 1.5 million (DHSS estimate 1 to 2 million), i.e. 2.5% of the population,
- the mean call duration is two minutes.

4.2 Interference model details

4.2.1 Mobile transmit power requirement

As discussed previously, it is a common strategy to operate mobiles at minimum power and therefore the transmitted power may be calculated from a knowledge of the radio link budget needed to sustain satisfactory quality calls from mobiles distributed over the cell. The link budget comprises cable loss, mobile antenna gain (0 dBi), interference margin (3 dB), log normal fading margin (5 dB) and Rayleigh fading margin (10 dB), a typical receiver noise figure (8 dB) and the bit rate (270.833 kbit/s) and temperature (290 K) of the system. The minimum signal level needed at the base-station receiver can be calculated from this information giving a receiver sensitivity of -94 dBm for a 10^{-2} BER.

To calculate the appropriate transmit power, the radio path loss to the three concentric rings from the base-station antenna has to be calculated; this is achieved using Hata's formulation of Okumura's propagation loss curves for different environments [5].

The path loss in dB is given by:

$$L_{path} = 69.55 + 26.16 \log f_c - 13.82 \log_{10} h_t - A(h_r) + (44.9 - 6.55 \log_{10} h_r) \log_{10} d_{path} \text{ (dB)}$$

where f_c is frequency in MHz

h_t is the height of the transmitter antenna height (62 m)

d_{path} is the distance from the base station in km,

and from equation 3.27 in Holbeche [6]:

$$A(h_r) = 3.2 (\log_{10} (11.75 h_r))^2 - 4.97 \text{ (dB)}$$

where h_r is the receive antenna height (2 m).

These equations thus simplify to give:

$$136 \quad L_{path} = 121 + 33 \log_{10} d_{path} \text{ (dB)}.$$

Hence the effective radiated power (ERP) needed by mobiles to transmit to the base station are:

- 500 mW (27 dBm) at 1 km
- 1.9 W (32.8 dBm) at 1.5 km
- 4.9 W (36.9 dBm) at 2 km

It is interesting to note that a class 4 (2 W) mobile would not be supported at a 2 km radius in this scenario, yet the major part of the current GSM phone population are class 4 mobiles, which are not coverage-limited in the London environment. This means that operators already use smaller cells or have higher gain antennas on the base station receivers than estimated. Therefore there will be less interference than predicted by the model.

4.2.2 Area of interference around each mobile

Each mobile will have an area around it in which a hearing aid user may receive interference. As discussed previously, this will not only depend on the distance from the mobile antenna, but also upon the attitude of both the mobile and the hearing aid wearer.

An equation for the interfering distance, d_{int} , may be derived from the following:

$$S = E^2 / 120\pi \text{ (power density in free space)}$$

$$\text{and } S = G p_t / (4 \pi d_{int}^2) \text{ (power density on surface of a sphere)}$$

where S = power density,
 G = antenna gain,
 E = field strength.

$$\text{Therefore } d_{int}^2 = G 30 P_t / E^2$$

As the antenna gain of the mobile is 1, G becomes redundant.

The laboratory tests of hearing aids [2], showed that 'equivalent susceptibility arcs' of 4V/m for 90 ° and 10V/m for the remaining 270 ° could be made (Fig 4). It can be

shown that this equates to a total area of susceptibility equal to $2.18 P_t Wm^2$

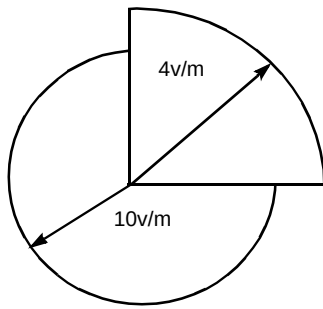


Fig 4 Equivalent susceptibility arcs (not to scale).

4.2.3 Number of interfering mobiles

If 5 MHz is utilised by each of the two UK 900 MHz GSM operators, this will lead to a total of fifty 200 KHz carriers being required with a maximum of eight traffic channels per carrier. Assuming a 'tight' three-cell cluster reuse pattern for cell layout (worse case interference), this gives $50/3 = 16$ carriers per cell (8 per operator). Sixteen of these channels will not be available, as they are used for cell broadcast control channels (BCCH). Therefore, there will be 112 traffic channels available. (Note: this paper does not consider interference from DSC1800 operators, which are likely to have a different cell architecture, powers and propagation characteristics.)

Using a 10-MHz allocation the number of mobiles on each of the concentric rings will be:

$$112 \times 1/4.5 = 25 \text{ mobiles at 1 km}$$

$$112 \times 1.5/4.5 = 37 \text{ mobiles at 1.5 km}$$

$$112 \times 2/4.5 = 50 \text{ mobiles at 2 km}$$

The affected area around each mobile is:

$$2.18 \times 0.5 = 1.1 \text{ m}^2$$

$$2.18 \times 1.9 = 4.1 \text{ m}^2$$

$$2.18 \times 4.9 = 10.7 \text{ m}^2$$

Thus the total affected area for all the mobiles is $(25 \times 1.1) + (37 \times 4.1) + (50 \times 10.7) = 714.2 \text{ m}^2$

This represents $714/1.26 \times 10^7 = 0.0056\%$ as a percentage of a 2 km radius cell.

4.3 Scenarios

To understand the impact of GSM interference within a cell, the model uses a number of scenarios in which hearing aid wearers may find themselves. These have been compiled from considerations of distances, e.g. distance from car to pavement, and the number of people who may be in a particular vicinity, like a railway station during the rush hour. This led to the following conclusions for cellular interference in the early 1990s (with 10 MHz allocated to GSM):

- it is unlikely that interference will be perceived from vehicles on the road while walking on the pavement,
- the probability of interference from a payphone on a train is 0.012,
- the probability of interference from hand-held mobiles in a railway station is 0.00154,
- the probability of interference from hand-held mobiles in an office block is 0.00016,
- interference from hand-held mobiles on a London street during peak time will cause 2.9 s of interference every 23 min,
- the probability of interference on a train is 0.009.

4.3.1 'Day in the life' scenarios

It is useful to extend the scenario study to 'day in the life' scenarios, e.g. a daily commuter from outside London or a retired person dwelling in a rural area, or perhaps a motorway traffic jam, in order to get a feel for how often and when a hearing aid wearer will experience interference.

This is shown in the example depicted in Table 1.

Table 1 Example 'day in the life' scenario.

Activity — daily commuter from outside London (10MHz GSM allocation)	Time taken	Interference
Travel from home (rural) to railway station and return	$2 \times 15 \text{ min} = 30 \text{ min}$	Very unlikely
Return train journey to London	$2 \times 1 \text{ hr} = 2 \text{ hr}$	$2 \times 2 \text{ min}$ every day
Time spent leaving and waiting for train	$2 \times 15 \text{ min} = 30 \text{ min}$	$1 \times 2 \text{ min}$ every two weeks
Tube journey	$2 \times 15 \text{ min} = 30 \text{ min}$	no coverage
Walking to and from office	$2 \times 15 \text{ min} = 30 \text{ min}$	$4 \times 3 \text{ sec}$ burst every day
Time spent in office	8 hr	$1 \times 2 \text{ min}$ every 9 days

There are obviously variables on this analysis, e.g. a large part of the population of hearing aid wearers may be retired and will not travel to London every day and will therefore receive interference less frequently and there tends to be a natural 'exclusion zone' around a person using a mobile telephone. However, the model provides a useful means of estimating the impact of interference from mobile telephones, the effect of changes in equipment immunity standards and the effect of GSM system developments.

4.4 *Maturity of GSM mobile systems and effect on interference*

The growth of cell-phone users with the UK has been dramatic within the past decade. Initially there were only analogue 'TACS' mobile phones, but a gradual changeover to GSM has been taking place since the early 1990s. This is accelerating now that technology has become cheaper and other issues of quality, security and additional data services are becoming important to customers. Operators have been faced with the challenge of accommodating an ever-increasing number of customers with the same spectrum allocation. The first step in achieving this has been to split existing cells into smaller ones, thus reducing the radii, the power needed from mobiles and hence the possibility of interference. Against this, there are more users, with a greater population of hand-held mobiles, which are often used within buildings, and two new operators in the 1800 MHz band who are laying down their networks. In order to provide coverage, rather than capacity, during initial 'roll-out', these new operators will make their cell sizes as large as possible in order to give ubiquitous coverage. This will give rise to a greater probability of interference.

However, it has been estimated that at present about 1 in 10 of the adult population have a mobile phone in the UK and this will grow to 1 in 4 by the turn of the century. This means that all networks will have to be mature and use very small cell sizes. This will actually reduce the probability of interference.

In addition, simple modifications to some types of hearing aid, such as coating the inside of the plastic case with conductive paint, can raise immunity from 3 V/m to 10 V/m. If this immunity was in place for all vital equipment, the probability of interference would be reduced dramatically. There would still be a problem for hearing aid users wanting to make use of GSM handsets, and this is currently being addressed within the ETSI GSM2 working group, although no product for the hearing impaired has emerged within the UK to date.

There is much discussion at present on what the ultimate 'penetration' of GSM phones will be; clearly people are

using phones within buildings and some operators have responded by installing micro-cells to provide coverage for office use. This will mean that mobiles will operate at their lowest power level and will be very unlikely to cause interference. Whether GSM phones, which are inherently low-data bandwidth, will meet the needs of a modern office is of great interest to those looking at so-called 3rd generation systems. The technology of the future could be a dual mode system such as GSM and DECT (digital European cordless telephony) or WLAN (wireless local area network). These options will also need to be examined for their interference potential.

5. Health fears

Health fears have sometimes been mixed with fears about interference to medical electronics, such as hearing aids and heart 'pacemakers' which is an electrical interference issue, although the fears and experience of those concerned are clearly just as important.

The issue can therefore be broadly broken into two main parts — exposure to radio waves and interference to medical electronics.

5.1 *Exposure to radio waves*

Like any radiation, e.g. sunlight, radio waves can affect health if too powerful. Safety guidelines for exposure to radio frequency radiation have been developed over many years of research and practical experience. These guidelines are based upon the only known and demonstrable effect, that of heating body tissue. Exposure levels are set well below those which would cause a discernible effect in parts of the body most susceptible, e.g. the human eye.

Although the safety levels vary between national bodies [7 — 10], the radiated power levels from the current generation of hand-held mobile phones will not exceed any of these guidelines if used normally. The most stringent standard assumes that the user will receive exposure for up to 24 hours a day.

There have been a number of reports in the press about athermal, e.g. carcinogenic or neural effects from using hand-held mobiles. Despite the high level of research activity in this area, and a substantial population of cellular radio users for more than 15 years in the USA, there is no conclusive evidence to support these reports. Nevertheless, research continues in this area, by bodies such as COST 244 (Biomedical Effects of Electromagnetic Fields).

5.2 Interference to medical electronics

The issue of hearing aids has already been discussed, but there are many other types of medical electronic equipment in use which may be susceptible to GSM interference.

The use of cellular mobiles can be forbidden in a controlled environment such as a hospital, where there may be old equipment of unknown immunity. Clearly this is a reasonable policy until tests are undertaken to assess this risk. However, the use of life saving equipment such as heart 'pacemakers', may be exposed to GSM transmission by the person with the pacemaker or by another person using a mobile nearby.

Tests have been performed on certain models of pacemakers using a 'phantom' (an imitation of a human body filled with saline solution) [11]. Some interference effect was noticed when the GSM transmitter was placed next to one type of pacemaker in air when operating at a high power level, but not when the pacemaker was within the phantom. This is obviously a reassuring result, but reinforces the need for manufacturers of critical medical electronics to make high equipment immunity a major design feature in new products.

6. Visual impact of base-station antennas

The visual impact of cellular base-station antennas has received much press of late. This may be partly because of the very large number of new base sites being acquired and commissioned in a very short space of time by all operators. For relatively new PCN operators, it is crucial to provide coverage comparable with the established operators quickly, in order to attract customers and make the network profitable.

Operators have to obtain permission from local councils for their base sites by submitting a 'general development order'. The installation must fall within statutory guidelines, which places limits on the installation, such as height, antenna size and appearance. Environmentally sensitive areas, e.g. areas of outstanding natural beauty, are treated as separate cases.

Having submitted a 'general development order', the council have 28 days in which to refuse planning permission. Public opposition can lead to an appeal being made directly to the UK Government 'Department of Environment', who also have a mandate to encourage the development of mobile communications. This situation is currently under review and it is expected that a general tightening up of restrictions will take place.

There is a conundrum between locating base sites away from visually sensitive areas where people like to be, and providing them adequate coverage. As discussed earlier, as cellular systems mature, there will be less need to transmit high power and antennas will be made intentionally lower to restrict coverage in densely populated areas for capacity reasons. This will make the antennas and towers less obvious on the skyline, perhaps no more obvious than a lamp standard. There are, however, always likely to be high towers and antennas in less densely populated and rural regions to achieve widespread coverage. The impact of these can be lessened by 'blending' and disguising antennas into existing structures whenever possible. Towers can also be located in areas where they are more acceptable among other clutter, such as on industrial estates or alongside railways. These solutions are often a compromise between providing effective coverage at a realistic cost quickly and seeking sub-optimal engineering solutions, which are readily acceptable to the public.

Some examples of towers are shown in Figs 5–8 below. These are neither the worst nor the best installations, but are used for illustrative purposes.

Figure 5 shows a typical antenna installation on top of a block of flats. It has the advantage of using an existing structure, suitable accommodation and power source, and can be maintained relatively easily.



Fig 5 High urban site.

Figure 6 shows how antennas have been 'blended into the building structure', by mounting sectorized antennas on the building corners; however, this makes maintenance more difficult.



Fig 6 Concealed urban site.

Figure 7 shows good use of an existing structure and the antennas (mounted on the legs of the tower) are not clearly visible a couple of hundred metres from the site.

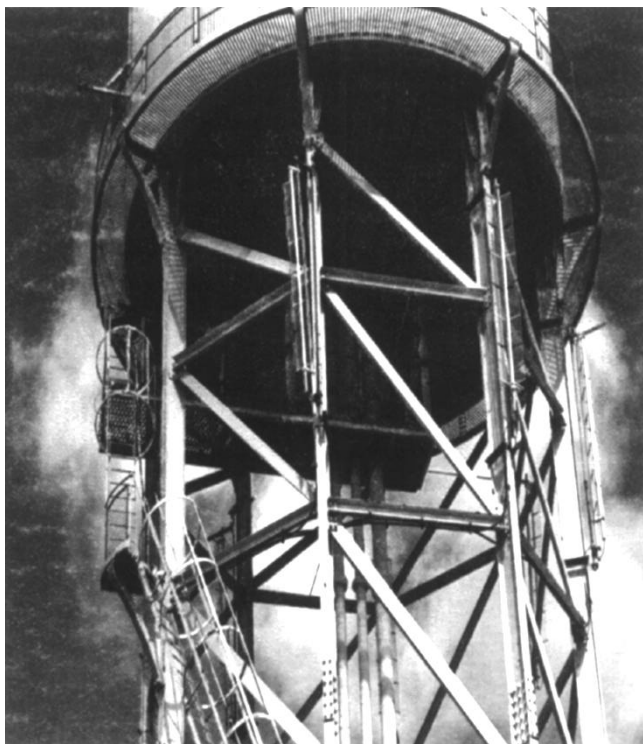


Fig 7 Existing structure — Harwich water tower (close up).

Figure 8 shows an example of how a small suburban cell 'pitchfork' antenna can be made to blend in with the general clutter of existing street furniture and trees.



Fig 8 Small suburban cell.

7. Conclusions

The rights of a radio system operator and those who receive interference has always attracted lively debate.

Although all radio systems are capable of causing interference, GSM presents particular EMC problems compared with other radio access techniques. While the new EU equipment immunity legislation will help to overcome this, a better immunity standard for critical electronic equipment would be of benefit. The effect of interference can be presented using simple graphical and subjective assessment techniques.

The BT interference model provides a useful way of analysing the impact of GSM and other radio system interference to other mobile or fixed electronic equipment within a cell. The results generated show a relatively low probability of people wearing hearing aids receiving interference.

There are a number of health fears, but no conclusive evidence exists to support them at present; the situation continues to be monitored.

Cellular network operators are increasing capacity quickly, which requires more base sites; this also means that mobile power will generally be lower and base sites will be less high, resulting in less radio interference and visual impact. The introduction of DCS1800 MHz networks run contrary to this argument in their early stages, but all networks will eventually operate at low power in outdoor urban environments, where interference may be a problem.

The evolution of mobiles into compact hand-held units means that they tend to be used within buildings more frequently, but building penetration losses means that the power will be increased. Building coverage in public and office buildings can be provided using special coverage techniques or other radio access technologies. This will give less interference, better performance for users and extra traffic for cellular operators, but interference may still be a problem in buildings without local radio access.

The visual impact of cellular antennas can be reduced by camouflage, installing sites in sub-optimal locations and by careful blending with building structures; however, the evolution of cellular into very high capacity networks is likely to reduce the visual impact of antennas. Gaining planning permission for new base-station sites may become more difficult in the future.

BT has been active in understanding the problem and working towards solutions. It is not an authority on health issues, but takes input from reputable bodies such as NRPB, which has not identified any specific hazards.

8. Ongoing work

BTL is continuing work on cellular interference issues through the 'Link Personal Communications' programme and proposed support to Cellnet on a GSM MoU group, set up to review EMC and bio-effects. The link programme, whose participants are CCL, ERA and the Universities of York and Hull, aims to 'reduce the occurrence of undesired electromagnetic interactions involving radio-based terminals'.

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